33kV INCOMING SUPPLY FROM F00-SW-010 PANEL B22
PLANT 33kV DISTRIBUTION SWITCHBOARD
2000-F00-SLD-EL-10006

F23-TR-016
33 / 11kV
5000kVA
Dyn11
z=7%
KNAN

F23-NR-016
FINAL TAILS PUMPS 33/11kV
TRANSFORMER F23-TR-016
NEUTRAL EARTHING RESISTOR
2000-EL011-B02-10004

30/0.05A
EFCT

25A FOR 10s

254Ω +/-5%

SENSING
ER-15kV

240V SUPPLY
F23-DB-001 SWITCHROOM L&SP DB
2000-F23-TLD-EL-10051

F23DB001-W6
F23DB001-BK6
HEATER

F23-SW-016
FINAL TAILS PUMPS 11kV SWITCHBOARD PANEL A04
2000-EL003-F16-10005

EARTH BAR

QCE

BAD

PFV1

BCF2
(PART OF)

CONSTRUCTION NOTE 1

BCT1 300 / 1A 2.5VA CL 0.2
BCT2 300 / 1A 2.5VA CL 0.2
BCT3 300 / 1A 2.5VA CL 0.2
SEL 735

BCT1 300 / 1A 2.5VA 5P20
BCT2 300 / 1A 2.5VA 5P20
BCT3 300 / 1A 2.5VA 5P20
SEL 751

2000-EL003-F16-10005
SHEET 16

-QAB
630A

RD
WH
BU

11kV 1250A 3-E 25kA FOR 3s

F23TR016-JB
FINAL TAILS PUMPS TRANSFORMER F23-TR-016 PROTECTION DEVICES TERMINAL BOX
2000-EL011-D02-10001

PRESSURE RELIEF VENT TRIP — PRV
OIL LEVEL ALARM (MINIMUM) — OIL LEVEL
OIL LEVEL ALARM (MAXIMUM) — OIL LEVEL
OIL TEMPERATURE ALARM (100°C) — OIL TEMP
OIL TEMPERATURE TRIP (110°C) — OIL TEMP
WINDING TEMPERATURE ALARM (110°C) — WINDING
WINDING TEMPERATURE TRIP (125°C) — WINDING

F23-SW-016
FINAL TAILS PUMPS 11kV SWITCHBOARD PANEL A04
2000-EL003-F16-10005

BCF1 (PART OF)

1535 / 1545 — X2 10 / 55 — 48VDC CONTROL — D10 IN409 — TRANSFORMER PROTECTION TRIP
1595 / 1605 — X2 14 / 59 — 48VDC CONTROL — D14 IN413 — TRANSFORMER OIL LEVEL MAXIMUM ALARM
1625 / 1635 — X2 15 / 60 — 48VDC CONTROL — D15 IN414 — TRANSFORMER OIL TEMPERATURE ALARM
1685 / 1695 — X2 16 / 61 — 48VDC CONTROL — A10 IN101 — TRANSFORMER WINDING TEMPERATURE ALARM

SHEET 13

F23SR001-LCS
SWITCHROOM EMERGENCY SHUTDOWN PUSHBUTTON (LOCATED IN AIR LOCK)
2000-F23-TER-EL-10202

TO 2000-F23-SCM-EL-10202 & 2000-F23-SCM-EL-10223

11 / 21 — 12 / 22 — E-SHUT

13 / 11 — 14 / 12 — E-SHUT
NOTE 5

1835 — X3 23
1855 — X3 33
SWITCHROOM EMERGENCY SHUTDOWN PUSHBUTTON CONNECTION TO THE VCB 1st TRIP CIRCUIT
SHEET 8

1955
1965

TO F00-SW-010 - PLANT 33kV HV DISTRIBUTION SWITCHBOARD FEEDER
2000-F00-TER-EL-10143

CONSTRUCTION NOTE 2

X5 41 / 40 — NOTE 4 — X5 32
X5 42 / 43 — NOTE 4 — X5 34

BCF1 (PART OF)
C03 OUT302
C04
PROTECTION TRIP UPSTREAM

F23-DCS-409
FINAL TAILS PUMPS 415V SWITCHROOM DCS PANEL
2000-F23-TER-IC-10014

X2 30 — F23SW016OP+48VDC — R1 11 / 1A — 1B — 01/03/01 — DCS REMOTE CB OPEN
X2 63 — F23SW016OP — R1 14
X2 31 — F23SW016CL+48VDC — R2 11 / 2A — 2B — 01/03/02 — DCS REMOTE CB CLOSE
X2 64 — F23SW016CL — R2 14

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F23SW016-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.
3. ALL ALARMS AND TRIPS SHALL BE FAIL-SAFE EXCEPT THE BUCHHOLZ RELAY. NOTE THAT THE PROTECTION RELAY INPUTS SHALL BE INVERTED, I.E. ALARM AND TRIP WHEN CONTACT IS OPENED.
4. NOTE THAT CROSS SECTIONAL AREA OF CABLES FROM F00-SW-010 ARE INCREASED IN SIZE - REFER TO THE CABLE SCHEDULE.
5. E-SHUT CIRCUIT IS FAIL SAFE. NORMALLY CLOSED CONTACTS WILL BE OPEN WHEN THE E-SHUT BUTTON IS UN-ACTIVATED AND THE CIRCUIT HEALTHY.

CONSTRUCTION NOTES

1. CONFIRM TERMINAL X4:23 IS SPARE.
   INSTALL WIRING FROM TERMINAL X4:23 TO (SE330 AU) BCF2 TERMINAL R:6.
2. INSTALL LINKS ACROSS SPARE TERMINALS X5:40-X5:41 AND X5:42-X5:43.
   INSTALL BRIDGING WIRE FROM TERMINAL X5:32 TO TERMINAL X5:40.
   INSTALL BRIDGING WIRE FROM TERMINAL X5:34 TO TERMINAL X5:42.

LEGEND

- xx/xx/xx — DCS DIGITAL INPUT
- xx/xx/xx — DCS DIGITAL OUTPUT
- xx/xx/xx — DCS ANALOGUE INPUT
- xx/xx/xx — DCS ANALOGUE OUTPUT
- xx/xx/xx — DCS ANALOGUE RTD INPUT
- 11kV SWITCHBOARD TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- (x) DEVICE TERMINAL

PANEL A04

2000-F23-SCM-EL-10107_Rev1

F23-SW-016 - F23-SW-016 incomer - schematic diagram